Exercise-1

Marked Questions can be used as Revision Questions.

PART - I : SUBJECTIVE QUESTIONS

SECTION (A) : PRINCIPLE OF SUPERPOSITION, PATH DIFFERENCE, WAVEFRONTS, AND COHERENCE

A-1. Two sources of intensity I & 4I are used in an interference experiment. Find the intensity at points where the waves from the two sources superimpose with a phase difference of **[REE 1991, 3]**

(a) zero (b) $\frac{\pi}{2}$ & (c) They meet at phase difference of π.

A-2. An electromagnetic wave travelling through a transparent medium is given by

$E_x(y,t) = E_{ox} \sin 2\pi\left[\frac{y}{5\times10^{-7}} - 3\times10^{14}t\right]$ in SI units. Then what is the refractive index of the medium?

SECTION (B) : YDSE WITH MONOCHROMATIC LIGHT

B-1. What is the effect on the fringe width of interference fringes in a Young's double slit experiment due to each of the following operations :
(a) The screen is moved away from the plane of the slits.
(b) The (monochromatic) source is replaced by another (monochromatic) source of shorter wavelength.
(c) The separation between the two slits is increased.
(d) The width of two slits are slightly increased.
[In each operation, take all parameters, other than the one specified to remain unchanged]

B-2. Two slits separated by a distance of 1 mm, are illuminated with red light of wavelength 6.5×10^{-7} m. The interference fringes are observed on a screen placed 1 m from the slits. Find the distance between the third dark fringe and the fifth bright fringe on the same side of the central maxima.

B-3. In a Young's double slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index (4/3), without disturbing the geometrical arrangement, what is the new fringe width?

B-4. Find the angular fringe width in a Young's double slits experiment with blue-green light of wavelength 6000 Å. The separation between the slits is 3.0×10^{-3} m.

SECTION (C) : YDSE WITH POLYCHROMATIC LIGHT

C-1. A source emitting two light waves of wavelengths 580 nm and 700 nm is used in a young's double slit interference experiment. The separation between the slits is 0.20 mm and the interference is observed on a screen placed at 150 cm from the slits. Find the linear separation between the first maximum (next to the central maximum) corresponding to the two wavelengths.

SECTION (D) : YDSE WITH GLASS SLAB, OPTICAL PATH

D-1. A flint glass and a crown glass are fitted on the two slits of a double slit apparatus. The thickness of the strips is 0.40 mm and the separation between the slits is 0.12 cm. The refractive index of flint glass and crown glass are 1.62 and 1.52 respectively for the light of wavelength 480 nm which is used in the experiment. The interference is observed on a screen a distance one meter away. (a) What would be the fringe-width? (b) At what distance from the geometrical centre will the nearest maximum be located?

D-2. Find the thickness of a plate which will produce a change in optical path equal to one fourth of the wavelength λ of the light passing through it normally. The refractive index of the plate is μ.

SECTION (E) : YDSE WITH OBLIQUE INCIDENCE AND OTHER MODIFICATIONS IN EXPERIMENTAL SETUP OF YDSE

E-1. A parallel beam of monochromatic light of wavelength λ is used in a Young's double slit experiment. The siits are separated by a distance d and the screen is placed parallel to the plane of the slits. The incident beam makes an angle $\theta = \sin^{-1}\left(\frac{\lambda}{2d}\right)$ with the normal to the plane of the slits. A transparent sheet of refractive index. 'μ' and thickness $t = \frac{\lambda}{2(\mu-1)}$ is introduced infront of one of the slit. Find the intensity at the geometrical centre.

SECTION (F) : THIN FILM INTERFERENCE

F-1. A soap film of thickness 0.3 μm appears dark when seen by the refracted light of wavelength 580 nm. What is the index of refraction of the soap solution, if it is known to be between 1.3 and 1.5?

F-2. A parallel beam of light of wavelength 560 nm falls on a thin film of oil (refractive index = 1.4). What should be the minimum thickness of the film so that it weakly transmits the light?

SECTION (G) : FOR JEE (MAIN)

G-1. A slit of width 'a' is illuminated by light of wavelength 6000 A^0 for what value of 'a' will the :
(i) First maximum fall at an angle of diffraction of 30° ?
(ii) First minimum fall at an angle of diffraction 30° ?

PART - II : ONLY ONE OPTION CORRECT TYPE

SECTION (A) : PRINCIPLE OF SUPERPOSITION, PATH DIFFERENCE, WAVEFRONTS, AND COHERENCE

A-1. Ratio of intensities of two light waves is given by 4 : 1. The ratio of the amplitudes of the waves is :
(A) 2 : 1 (B) 1 : 2 (C) 4 : 1 (D) 1 : 4

A-2. Two coherent monochromatic light beams of intensities I and 4I are superimposed; the maximum and minimum possible intensities in the resulting beam are :
(A) 5I and I (B) 5I and 3I (C) 9I and I (D) 9I and 3I

SECTION (B) : YDSE WITH MONOCHROMATIC LIGHT

B-1. The contrast in the fringes in any interference pattern depends on :
(A) Fringe width (B) Wavelength
(C) Intensity ratio of the sources (D) Distance between the sources

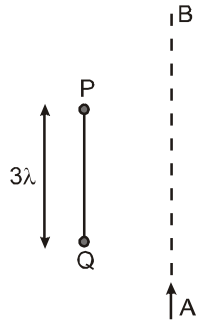
B-2. Initially interference is observed with the entire experimental set up inside a chamber filled with air. Now the chamber is evacuated. With the same source of light used, a careful observer will find that
(A) The interference pattern is almost absent as it is very much diffused
(B) There is no change in the interference pattern
(C) The fringe width is slightly decreased
(D) The fringe width is slightly increased

B-3. Yellow light emitted by sodium lamp in Young's double slit experiment is replaced by monochromatic blue light of the same intensity :
(A) fringe width will decrease. (B) fringe width will increase.
(C) fringe width will remain unchanged. (D) fringes will become less intense.

B-4. In a YDSE: D = 1 m, d = 1 mm and λ = 500 nm. The distance of 1000th maxima from the central maxima is:
(A) 0.5 m (B) 0.577 m (C) 0.495 m (D) does not exist

B-5. In a Young's double slit experiment, d = 1 mm, λ = 6000 Å & D = 1 m. The slits produce same intensity on the screen. The minimum distance between two points on the screen having 75 % intensity of the maximum intensity is:
(A) 0.45 mm (B) 0.40 mm (C) 0.30 mm (D) 0.20mm

B-6. Two coherent light sources each of wavelength λ are separated by a distance 3λ. The total number of minima formed on line AB which runs from $-\infty$ to $+\infty$ is:



(A) 2 (B) 4 (C) 6 (D) 8

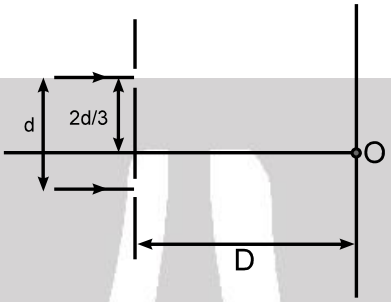
B-7. In a Young's double slit experiment the intensity at a point I where the corresponding path difference is one sixth of the wavelength of light used. If I_0 denotes the maximum intensity, the ratio $\frac{I}{I_0}$ is equal to

[Olympiad 2015 (stage-1)]

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{3}{4}$

SECTION (C) : YDSE WITH POLYCHROMATIC LIGHT

C-1. In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the nearest white spot on the screen from O is: [assume d << D, λ << d]



(A) 0 (B) d/2 (C) d/3 (D) d/6

C-2. The Young's double slit experiment is performed with blue and with green light of wavelengths 4360 Å and 5460 Å respectively. If X is the distance of 4th maximum from the central one, then :

(A) X(blue) = X(green) (B) X(blue) > X(green) (C) X(blue) < X(green) (D) $\frac{X(blue)}{X(green)} = \frac{5460}{4360}$

SECTION (D) : YDSE WITH GLASS SLAB, OPTICAL PATH

D-1. A two slit Young's interference experiment is done with monochromatic light of wavelength 6000 Å. The slits are 2 mm apart. The fringes are observed on a screen placed 10 cm away from the slits. Now a transparent plate of thickness 0.5 mm is placed in front of one of the slits and it is found that the interference pattern shifts by 5 mm. The refractive index of the transparent plate is : **[REE 1985]**

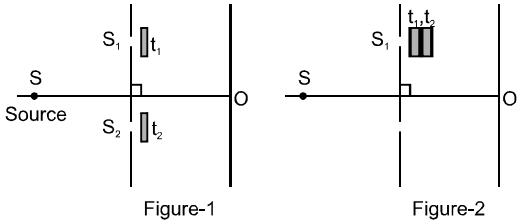
(A) 1.2 (B) 0.6 (C) 2.4 (D) 1.5

D-2. In a YDSE both slits produce equal intensities on the screen. A 100 % transparent thin film is placed in front of one of the slits. Now the intensity of the geometrical centre of system on the screen becomes 75 % of the previous intensity. The wavelength of the light is 6000Å and μ_{glass} = 1.5. The thickness of the film cannot be:

(A) 0.2 μm (B) 1.0 μm (C) 1.4 μm (D) 1.6 μm

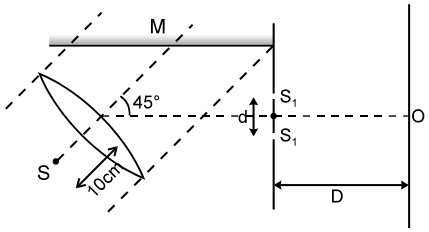
SECTION(E) : YDSE WITH OBLIQUE INCIDENCE AND OTHER MODIFICATIONS IN EXPERIMENTAL SETUP OF YDSE

E-1. In a YDSE experiment, thin films of thickness t_1 and t_2 are placed infront of slits S_1 and S_2 as shown in figure-1 and figure-2. It is observed that first minima and second maxima are produced at point 'O' in first and second experiment respectively. Point 'O' and 'S' are symmetrical with respect to S_1 and S_2. Both the films have same refractive index if $\frac{t_2}{t_1} = \frac{x}{25}$, then calculate 'x' :



(A) 3 (B) 9 (C) 12 (D) 15

E-2. In Youngs double slit experiment, distance between the slits is d and that between the slits and screen is D. Angle between principle axis of lens and perpendicular bisector of S_1 and S_2 is 45°. The point source S is placed at the focus of lens and aperture of lens is much larger than d. Assuming only the reflected light from plane mirror M is incident on slits, distance of central maxima from O will be :



(A) D (B) $\frac{D}{\sqrt{3}}$ (C) $D\sqrt{3}$ (D) $\frac{D}{\sqrt{4}}$

SECTION (F) : THIN FILM INTERFERENCE

F-1. White light is incident normally on a glass plate (in air) of thickness 500 nm and refractive index of 1.5. The wavelength (in nm) in the visible region (400 nm - 700nm) that is strongly reflected by the plate is:
(A) 450 (B) 600 (C) 400 (D) 500

SECTION (G) : FOR JEE (MAIN)

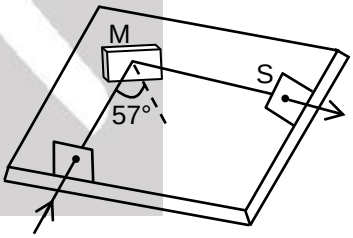
G-1. A slit of size 0.15 cm is placed at 2.1 m from a screen. On illuminated it by a light of wavelength 5×10^{-5} cm. The width of diffraction pattern will be:-
(A) 70 mm (B) 0.14 mm (C) 1.4 cm (D) 0.14 cm

G-2. The diameter of objective of a telescope is 1m. Its resolving limit for the light of wave length 4538 Å, will be
(A) 5.54×10^{-7} rad (B) 2.54×10^{-4} rad (C) 6.54×10^{-7} rad (D) None of these

G-3. When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is :
(A) $\frac{1}{2}I_0$ (B) $\frac{1}{4}I_0$ (C) zero (D) I_0

G-4. A single slit diffraction pattern is obtained using a beam of red light. What happens if the red light is replaced by blue light?
(A) There is no change in the diffraction pattern
(B) Diffraction fringes become narrower and crowded together
(C) Diffraction fringes become boarder and crowded together
(D) The diffraction pattern disappears

G-5. Shows a glass plate placed vertically on a horizontal table with a beam of unpolarised light falling on its surface at 57° with the normal. The electric vectors in the reflected light on the screen S will vibrate with respect to the plane of incidence :
(A) in a vertical plane
(B) in a horizontal plane
(C) in a plane making an angle of 45° with the vertical
(D) in a plane making an angle of 57° with the horizontal



G-6. Two point white dots are 1 mm apart on a black paper. They are viewed by eye of pupil diameter 3 mm. Approximately, what is the maximum distance at which these dots can be resolved by the eye?
[Take wavelength of light = 500 nm]
(A) 6m (B) 3m (C) 5m (D) 1m

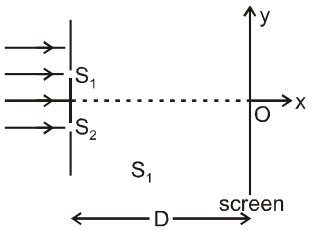
G-7. Visible light passing through a circular hole form a diffraction disc of radius 0.1 mm on a screen. If X-ray is passed through the same set-up, the radius of the diffraction disc will be :
(A) zero (B) < 0.1 mm (C) 0.1 mm (D) > 0.1 mm

G-8. The resolving power of a telescope is more when its objective lens has
(A) greater focal length (B) smaller focal length (C) greater diameter (D) smaller diameter

G-9. Resolving power of a microscope depends upon
(A) the focal length and aperture of the eye lens (B) the focal lengths of the objective and the eye lens
(C) the apertures of the objective and the eye lens (D) the wavelength of light illuminating the object

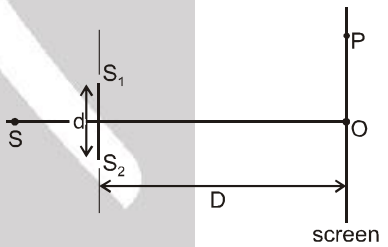
PART - III : MATCH THE COLUMN

1. A monochromatic parallel beam of light of wavelength λ is incident normally on the plane containing slits S_1 and S_2. The slits are of unequal width such that intensity only due to one slit on screen is four times that only due to the other slit. The screen is placed perpendicular to x-axis as shown. The distance between slits is d and that between screen and slit is D. Match the statements in column-I with results in column-II. ($S_1S_2 << D$ and $\lambda << S_1S_2$)



	Column-I	**Column-II**
(A)	The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{1}{9}$ th of maximum intensity.	(p) $\frac{D\lambda}{3d}$
(B)	The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{3}{9}$ th of maximum intensity.	(q) $\frac{D\lambda}{d}$
(C)	The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{5}{9}$ th of maximum intensity.	(r) $\frac{2D\lambda}{d}$
(D)	The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{7}{9}$ th of maximum intensity.	(s) $\frac{3D\lambda}{d}$

2. In a typical Young's double slit experiment, S_1 and S_2 are identical slits and equidistant from a point monochromatic source S of light having wavelength λ. The distance between slits is represented by d and that between slits and screen is represented by D. P is a fixed point on the screen at a distance $y = \frac{\lambda D_o}{2d_o}$ from central order bright on the screen: where D_o, d_o are initial values of D and d respectively. In each statement of column-I some changes are made to above mentioned situation.



The distance between the slits and the source is very large. The effect of corresponding changes is given in column-II. Match the statements in column-I with resulting changes in column-II.

	Column-I	**Column-II**
(A)	The distance d between the slits is doubled keeping distance between slits and screen fixed	(p) fringe width increases.
(B)	The distance D between slit and screen is doubled by shifting screen to right	(q) Magnitude of optical path difference between interfering waves at P will decrease.
(C)	The width of slit S_1 is decreased (such that intensity of light due to slit S_1 on screen decreases) and the distance D between slit and screen is doubled by shifting screen to right	(r) Magnitude of optical path difference between interfering waves at P will increase.
(D)	The whole setup is submerged in water of refractive index 4/3. (Neglecting absorption in medium)	(s) The intensity at P will increase

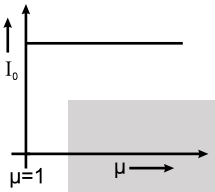
Exercise-2

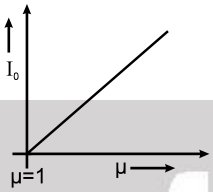
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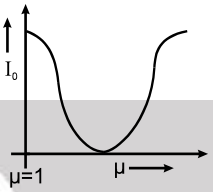
PART - I : ONLY ONE OPTION CORRECT TYPE

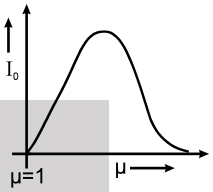
1. If the ratio of the intensity of two coherent sources is 4 then the visibility $[(I_{max} - I_{min})/(I_{max} + I_{min})]$ of the fringes is
(A) 4 (B) 4/5 (C) 3/5 (D) 9

2. In a YDSE experiment if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with 'μ' will be best represented by (μ ≥ 1). [Assume slits of equal width and there is no absorption by slab]

(A) 

(B) 

(C) 

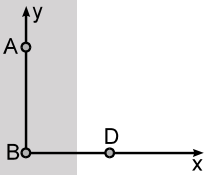
(D) 

3. In a Young's double slit experiment the slit is illuminated by a source having two wavelengths of 400 nm and 600 nm. If distance between slits, d = 1mm, and distance between the plane of the slit and screen, D = 10 m then the smallest distance from the central maximum where there is complete darkness is :
(A) 2mm (B) 3mm (C) 12 mm (D) there is no such point

4. If the first minima in a Young's slit experiment occurs directly infornt of one of the slits. (Distance between slit & screen D = 12 cm and distance between slits d = 5 cm) then the wavelength of the radiation used is :

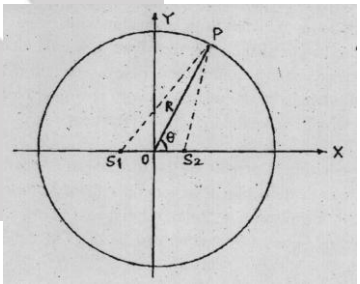
(A) 2 cm only (B) 4 cm only (C) 2m, $\frac{2}{3}$ cm, $\frac{2}{5}$ cm (D) 4cm, $\frac{4}{3}$ cm, $\frac{4}{5}$ cm

5. An interference is observed due to two coherent sources 'A' & 'B' having phase constant zero separated by a distance 4 λ along the y-axis where λ is the wavelength of the source. A detector D is moved on the positive x-axis. The number of points on the x-axis excluding the points, x = 0 & x = ∞ at which maximum will be observed is



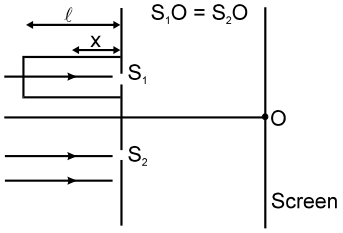
(A) three (B) four (C) two (D) infinite

6. Two coherent sources of light S_1 and S_2, equidistant from the origin, are separated by a distance 2λ as shown. They emit light of wavelength λ. Interference is observed on a screen placed along the circle of large radius R. Point is seen to be a point of constructive interference. Then angle θ (other than 0° and 90°) is



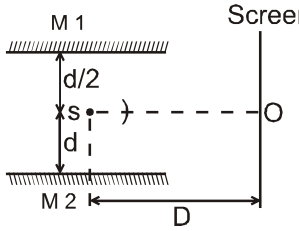
(A) 45°
(B) 30°
(C) 60°
(D) Not possible in the first quandrant

7. In the figure shown, a parallel beam of light is incident on the plane of the slits of a Young's double slit experiment. Light incident on the slit, S_1 passes through a medium of variable refractive index μ = 1 + ax (where 'x' is the distance from the plane of slits as shown), upto a distance 'ℓ' before falling on S_1. Rest of the space is filled with air. If at 'O' a minima is formed, then the minimum value of the positive constant a (in terms of ℓ and wavelength 'λ' in air) is :



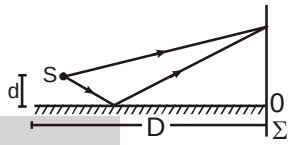
(A) $\frac{\lambda}{\ell}$ (B) $\frac{\lambda}{\ell^2}$ (C) $\frac{\ell^2}{\lambda}$ (D) None of these

8. M_1 and M_2 are two plane mirrors which are kept parallel to each other as shown. There is a point 'O' on perpendicular screen just infront of 'S'. What should be the wavelength of light coming from monchromatic source 'S'. So that a maxima is formed at 'O' due to interference of reflected light from both the mirrors. [Consider only 1st reflection]. [D > > d, d > > λ]



(A) $\frac{3d^2}{D}$ (B) $\frac{3d^2}{2D}$ (C) $\frac{d^2}{D}$ (D) $\frac{2d^2}{D}$

9. A long narrow horizontal slit lies 1 mm above a plane mirror. The interference pattern produced by the slit and its image is viewed on a screen Σ distant 1m from the slit. The wavelength of light is 600 nm. Then the distance of the first maxima above the mirror is equal to (d << D):



(A) 0.30 mm (B) 0.15 mm (C) 60 mm (D) 7.5 mm

10. A parallel monochromatic beam of light is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of the incident beam. At the first minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of the slit is: [Diffraction – Not in JEE syllabus now] **[JEE 1998, 2]**

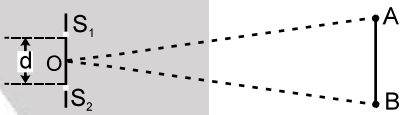
(A) 0 (B) π/2 (C) π (D) 2 π

11. A slit of width a is illuminated by parallel monochromatic light of wavelength λ. The value of a at which the first minimum of the diffraction pattern will from at θ = 30° is **[Olympiad (State-1) 2017]**

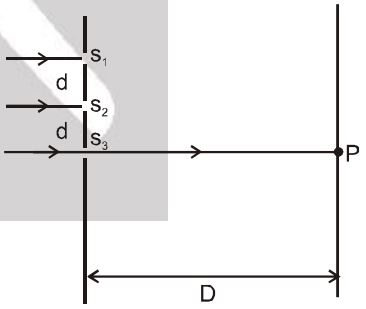
(A) λ/2 (B) λ (C) 2λ (D) 3λ

PART - II : SINGLE AND DOUBLE VALUE INTEGER TYPE

1. Figure shows two coherent sources S_1–S_2 vibrating in same phase. AB is a straight wire lying at a far distance from the sources S_1 and S_2. Let $\frac{\lambda}{d} = 10^{-3}$. ∠ BOA = 0.12°. How many bright spots will be seen on the wire, including points A and B.



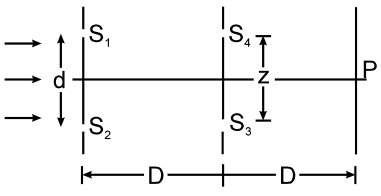
2. In the figure shown three slits s_1, s_2 and s_3 are illuminated with light of wavelength λ. λ << d and D >> d. Each slit produces same intensity I on the screen. If resultant intensity at the point on screen directly infront of s_2 is 3I then the maximum value of λ is $\frac{nd^2}{2D}$



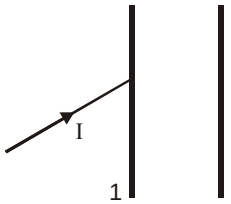
(i) Find value of n.

(ii) Also find intensity at point P if λ is of part (i)

3. Parallel monochromatic beam is falling normally on two slits S_1 and S_2 separated by d as shown in figure. By some mechanism, the separation between the slits S_3 and S_4 can be changed. The intensity is measured at the point P which is at the common perpendicular bisector of S_1S_2 and S_3S_4. When $z = \frac{D\lambda}{2d}$, the intensity measured at P is I and when $z = \frac{4D\lambda}{d}$, Intensity is xI. Find x.



4. A narrow monochromatic beam of light of intensity I is incident on a glass plate as shown in figure. Another identical glass plate is kept close to the first one & parallel to it. Each glass plate reflects 25 % of the light incident on it & transmits the remaining. The ratio of the minimum & the maximum intensities in the interference pattern formed by the two beams obtained after one reflection at each plate is 1 : n find value of n. **[JEE 1990, 7]**

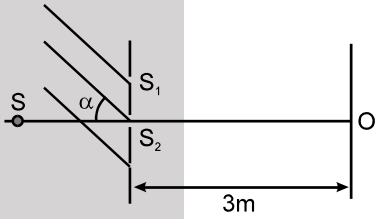


5. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength 5400 Å. It is found that the point P on the screen where the central maximum fell before the glass plates were inserted now has $(3/4)^{th}$ the original intensity. It is further observed that what used to be the 5th maximum earlier, lies below the point O while the 6th minimum lies above O. The thickness of the glass plate is $n \times 10^{-7}$m. Find the value of n. (Absorption of light by glass plate may be neglected) **[JEE 1997, 5/100]**

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

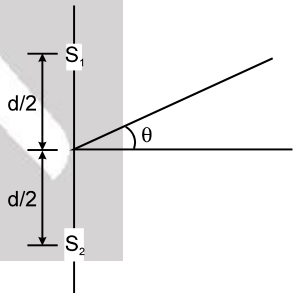
1. In Young's double slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes as 9. This implies : **[JEE 1982]**
(A) the intensities at the screen due to the two slits are 5 and 4 units
(B) the intensities at the screen due to the two slits are 4 and 1 units
(C) the amplitude ratio of the individual waves is 3
(D) the amplitude ratio of the individual waves is 2

2. A parallel beam of light (λ = 5000 Å) is incident at an angle α = 30° with the normal to the slit plane in a young's double slit experiment. Assume that the intensity due to each slit at any point on the screen is I_0. Point O is equidistant from S_1 & S_2.The distance between slits is 1mm.
(A) the intensity at O is $4I_0$
(B) the intensity at O is zero
(C) the intensity at a point on the screen 4m from O is $4I_0$
(D) the intensity at a point on the screen 4m from O is zero



3. A Young's double slit experiment is performed with white light:
(A) The maxima next to the central will be red. (B) The central maxima will be white
(C) The maxima next to the central will be violet (D) There will not be a completely dark fringe.

4. In an interfrence arrangement similar to Young's double-slit experiment, the slits S_1 & S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance d = 150.0 m and screen is at very large distance from slits. The intensity $I(\theta)$ is measured as a function of θ, where θ is defined as shown. Screen is at a large distance. If I_0 is the maximum intensity then I (θ) for $0 \le \theta \le 90^o$ is given by: **[JEE 1995, 1 + 2]**



(A) $I(\theta) = \frac{I_0}{2}$ for $\theta = 30^o$ (B) $I(\theta) = \frac{I_0}{4}$ for $\theta = 90^o$
(C) $I(\theta) = I_0$ for $\theta = 0^o$ (D) $I(\theta)$ is constant for all values of θ.

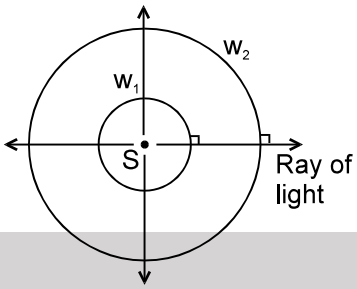
5. White light is used to illuminate the two silts in a Young's double slit experiment. The separation between the slits is b and the screen is at a distance d (>> b) from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are : **[JEE 1984]**

(A) $\lambda = \frac{b^2}{d}$ (B) $\lambda = \frac{2b^2}{d}$ (C) $\lambda = \frac{b^2}{3d}$ (D) $\lambda = \frac{2b^2}{3d}$

PART - IV : COMPREHENSION

Comprehension-1

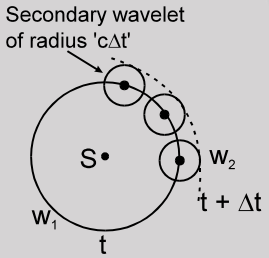
Huygen was the first scientist who proposed the idea of wave theory of light. He said that the light propagates in form of wavefronts. A wavefront is an imaginary surface at every point of which waves are in the same phase. For example the wavefronts for a point source of light is collection of concentric spheres which have centre at the origin. w_1 is a wavefront. w_2 is another wavefront.



The radius of the wavefront at time 't' is 'ct' in this case where 'c' is the speed of light. The direction of propagation of light is perpendicular to the surface of the wavefront. The wavefronts are plane wavefronts in case of a parallel beam of light.



Huygen also said that every point of the wavefront acts as the source of secondary wavelets. The tangent drawn to all secondary wavelets at a time is the new wavefront at that time. The wavelets are to be considered only in the forward direction (i.e. the direction of propagation of light) and not in the reverse direction. If a wavefront w_1 at time t is given, then to draw the wavefront at time $t + \Delta t$ take some points on the wavefront w_1 and draw spheres of radius '$c\Delta t$'. They are called secondary wavelets.

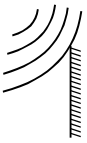


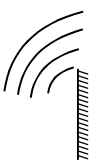
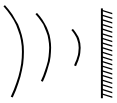
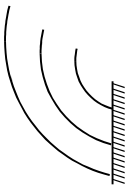
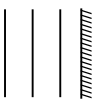
Draw a surface w_2 which is tangential to all these secondary wavelets. w_2 is the wavefront at time '$t + \Delta t$'. Huygen proved the laws of reflection and laws of refraction using concept of wavefronts.

1. A point source of light is placed at origin, in air. The equation of wave front of the wave at time t, emitted by source at t = 0, is (take refractive index of air as 1)

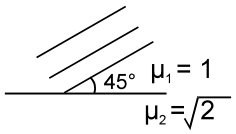
(A) $x + y + z = ct$ (B) $x^2 + y^2 + z^2 = t^2$ (C) $xy + yz + zx = c^2 t^2$ (D) $x^2 + y^2 + z^2 = c^2 t^2$

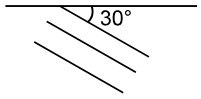
2. Spherical wave fronts shown in figure, strike a plane mirror. Reflected wave fronts will be as shown in

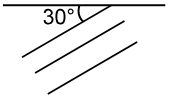


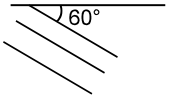
(A)  (B)  (C)  (D) 

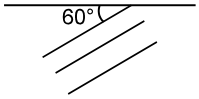
3. Wavefronts incident on an interface between the media are shown in the figure. The refracted wavefronts will be as shown in



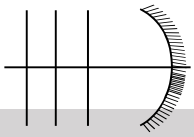
(A) 

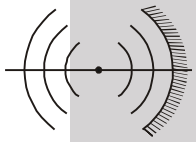
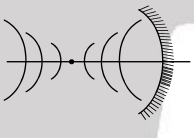
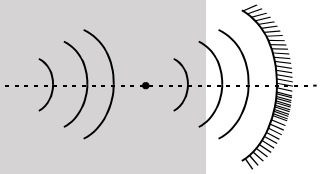
(B) 

(C) 

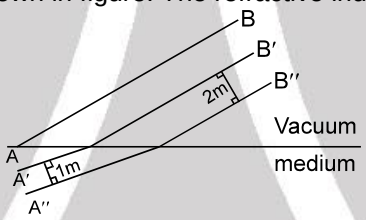
(D) 

4. Plane wavefronts are incident on a spherical mirror as shown. The reflected wavefronts will be



(A) 
(B) 
(C) 
(D) 

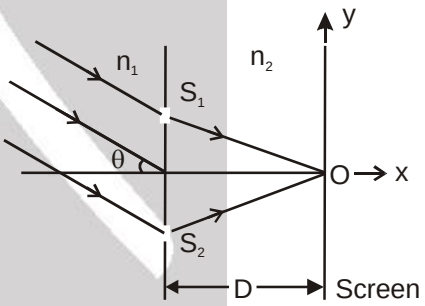
5. Certain plane wavefronts are shown in figure. The refractive index of medium is



(A) 2 (B) 4 (C) 1.5 (D) Cannot be determined

Comprehension-2

In the figure an arrangement of young's double slit experiment is shown. A parallel beam of light of wavelength 'λ' (in medium n_1) is incident at an angle 'θ' as shown. Distance $S_1O = S_2O$. Point 'O' is the origin of the coordinate system. The medium on the left and right side of the plane of slits has refractive index n_1 and n_2 respectively. Distance between the slits is d. The distance between the screen and the plane of slits is D. Using D = 1m, d = 1mm, θ = 30°, λ = 0.3mm, $n_1 = \frac{4}{3}$, $n_2 = \frac{10}{9}$, answer the following



6. The y-coordinate of the point where the total phase difference between the interefering waves is zero, is

(A) y = 0 (B) $y = +\frac{3}{4}$ m (C) $y = -\frac{3}{4}$ m (D) $-\frac{1}{\sqrt{3}}$ m

7. If the intensity due to each light wave at point 'O' is I_0 then the resultant intensity at point 'O' will be -

(A) Zero (B) $2I_0\left(1+\cos\frac{40\pi}{9}\right)$ (C) $3I_0$ (D) I_0

8. y-coordinate of the nearest maxima above 'O' will be -

(A) $\frac{150}{\sqrt{154}}$ cm (B) 24 cm (C) $\frac{100}{\sqrt{99}}$ cm (D) None of these

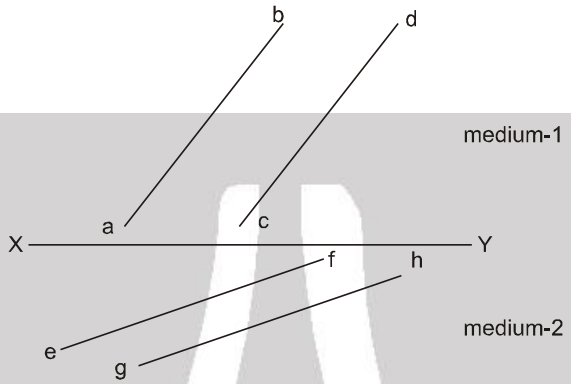
Exercise-3

✍ Marked Questions can be used as Revision Questions.
*** Marked Questions may have more than one correct option.**

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

Paragraph of question no. 1 to 3

The figure shows surface XY separating two transparent media, medium–1 and medium–2. The lines ab and cd represent wavefronts of a light wave travelling in medium–1 and incident on XY. The lines ef and gh represent wavefronts of the light wave in medium–2 after refraction. **[JEE 2007, 4+4+4/162]**



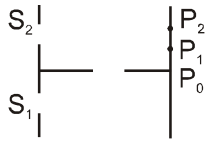
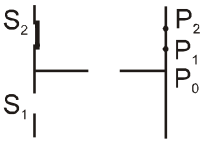
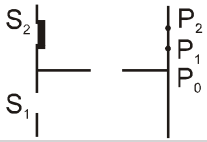
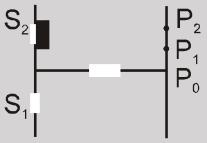
1. Light travels as a
(A) parallel beam in each medium
(B) convergent beam in each medium
(C) divergent beam in each medium
(D) divergent beam in one medium and convergent beam in the other medium

2. The phases of the light wave at c, d, e and f are ϕ_c, ϕ_d, ϕ_e and ϕ_f respectively. It is given that $\phi_c \neq \phi_f$:
(A) ϕ_c cannot be equal to ϕ_d (B) ϕ_d can be equal to ϕ_e
(C) $(\phi_d - \phi_f)$ is equal to $(\phi_c - \phi_e)$ (D) $(\phi_d - \phi_c)$ is not equal to $(\phi_f - \phi_e)$

3. Speed of light is
(A) the same in medium–1 and medium–2 (B) larger in medium–1 than in medium–2
(C) larger in medium–2 than in medium–1 (D) different at b and d

4*.✍ In a Young's double slit experiment, the separation between the two slits is d and the wavelength of the light is λ. The intensity of light falling on slit 1 is four times the intensity of light falling on slit 2. Choose the correct choice(s). **[JEE 2008, 3/163]**
(A) If $d = \lambda$, the screen will contain only one maximum
(B) If $\lambda < d < 2\lambda$, at least one more maximum (besides the central maximum) will be observed on the screen
(C) If the intensity of light falling on slit 1 is reduced so that it becomes equal to that of slit 2, the intensities of the observed dark and bright fringes will increase
(D) If the intensity of light falling on slit 2 is increased so that it becomes equal to that of slit 1, the intensities of the observed dark and bright fringes will increase

5. **Column I** shows four situations of standard Young's double slit arrangement with the screen placed far away from the slits S_1 and S_2. In each of these cases $S_1P_0 = S_2P_0$, $S_1P_1 - S_2P_1 = \lambda/4$ and $S_1P_2 - S_2P_2 = \lambda/3$, where λ is the wavelength of the light used. In the cases B, C and D, a transparent sheet of refractive index μ and thickness t is pasted on slit S_2. The thicknesses of the sheets are different in different cases. The phase difference between the light waves reaching a point P on the screen from the two slits is denoted by $\delta(P)$ and the intensity by $I(P)$. Match each situation given in **Column-I** with the statement(s) in **Column-II** valid for that situation. **[JEE 2009, 8/240]**

Column–I		Column–II	
(A)		(p)	$\delta(P_0) = 0$
(B)	$(\mu - 1)t = \lambda/4$ 	(q)	$\delta(P_1) = 0$
(C)	$(\mu - 1)t = \lambda/2$ 	(r)	$I(P_1) = 0$
(D)	$(\mu - 1)t = 3\lambda/4$ 	(s)	$I(P_0) > I(P_1)$
		(t)	$I(P_2) > I(P_1)$

6. Young's double slit experiment is carried out by using green, red and blue light, one color at time. The fringe widths recorded are β_G, β_R and β_B, respectively. Then **[IIT-JEE-2012, Paper-1; 3/70, –1]**

(A) $\beta_G > \beta_B > \beta_R$ (B) $\beta_B > \beta_G > \beta_R$ (C) $\beta_R > \beta_B > \beta_G$ (D) $\beta_R > \beta_G > \beta_B$

7. In the Young's double slit experiment using a monochromatic light of wavelength λ, the path difference (in terms of an integer n) corresponding to any point having half the peak intensity is : **[JEE (Advanced) 2013 ; P-1, 2/60]**

(A) $(2n+1)\frac{\lambda}{2}$ (B) $(2n+1)\frac{\lambda}{4}$ (C) $(2n+1)\frac{\lambda}{8}$ (D) $(2n+1)\frac{\lambda}{16}$

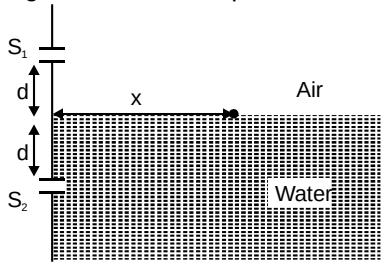
8. Using the expression $2d \sin\theta = \lambda$, one calculates the values of d by measuring the corresponding angles θ in the range 0 to 90º. The wavelength λ is exactly knowns and the error in θ is constant for all values of θ. As θ increases from 0º : **[JEE (Advanced) 2013 ; P-2, 3/60, –1]**

(A) the absolute error in d remains constant. (B) the absolute error in d increases.
(C) the fractional error in d remains constant. (D) the fractional error in d decreases.

9*. A light source, which emits two wavelengths $\lambda_1 = 400$ nm and $\lambda_2 = 600$ nm, is used in a Young's double slit experiment. If recorded fringe widths for λ_1 and λ_2 are β_1 and β_2 and the number of fringes for them within a distance y on one side of the central maximum are m_1 and m_2, respectively, then **[JEE (Advanced) 2014, P-1, 3/60]**

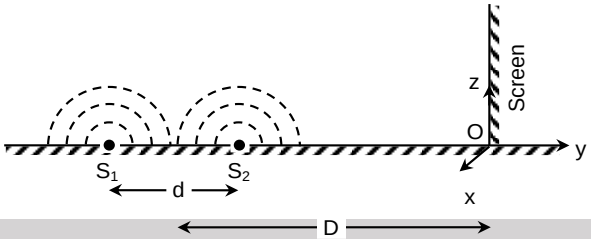
(A) $\beta_2 > \beta_1$
(B) $m_1 > m_2$
(C) From the central maximum, 3rd maximum of λ_2 overlaps with 5th minimum of λ_1
(D) The angular separation of fringes for λ_1 is greater than λ_2

10. A young's double slit interference arrangement with slits S_1 and S_2 is immersed in water (refractive index = 4/3) as shown in the figure. The positions of maxima on the surface of water are given by $x^2 = p^2m^2\lambda^2 - d^2$, where λ is the wavelength of light in air (refractive index = 1), 2d is the separation between the slits and m is an integer. The value of p is **[JEE (Advanced) 2015 ; P-1, 4/88]**



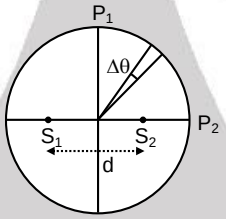
11*. While conducting the Young's double slit experiment, a student replaced to two slits with a large opaque plate in the x-y plane containing two small holes that act as two coherent point sources (S_1, S_2) emitting light of wavelength 600 nm. The student mistakenly placed the screen parallel to the x-z plane (for z > 0) at a distance D = 3 m from the mid-point of S_1S_2, as shown schematically in the figure. The distance between the sources d = 0.6003 mm. The origin O is at the intersection of the screen and the line joining S_1S_2. Which of the following is(are) true of the intensity pattern on the screen ?

[JEE (Advanced) 2016, P-2, 4/62, –2]



(A) Semi circular bright and dark bands centered at point O
(B) Hyperbolic bright and dark bands with foci symmetrically placed about O in the x-direction
(C) The region very close to the point O will be dark
(D) Straight bright and dark bands parallel to the x-axis

12.* Two coherent monochromatic point sources S_1 and S_2 of wavelength λ = 600 nm are placed symmetrically on either side of the center of the circle as shown. The sources are separated by a distance d = 1.8 mm. This arrangement produces interference fringes visible as alternate bright and dark spots on the circumference of the circle. The angular separation between two consecutive bright spots is $\Delta\theta$. Which of the following options is/are correct ? **[JEE (Advanced) 2017, P-2, 4/61, –2]**



(A) The total number of fringes produced between P_1 and P_2 in the first quadrant is close to 3000
(B) A dark spot will be formed at the point P_2
(C) At P_2 the order of the fringe will be maximum
(D) The angular separation between two consecutive bright spots decreases as we move from P_1 to P_2 along the first quadrant

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I. If I_0 denotes the maximum intensity, I/I_0 is equal to:

[AIEEE 2007 ; 3/120, –1]

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

Direction : Questions number 2 – 4 are based on the following paragraph.

An initially parallel cylindrical beam travels in a medium of refractive index $\mu(I) = \mu_0 + \mu_2 I$, where μ_0 and μ_2 are positive constants and I is the intensity of the light beam. The intensity of the beam is decreasing with increasing radius. **[AIEEE-2010, 4/144, – 1]**

2. How does beam travell after entering in medium ?
(1) diverge
(2) converge
(3) diverge near the axis and converge near the periphery
(4) travel as a cylindrical beam

3. The initial shape of the wavefront of the beam is :
(1) convex
(2) concave
(3) convex near the axis and concave near the periphery
(4) planar

4. The speed of light in the medium is
(1) minimum on the axis of the beam (2) the same everywhere in the beam
(3) directly proportional to the intensity I (4) maximum on the axis of the beam

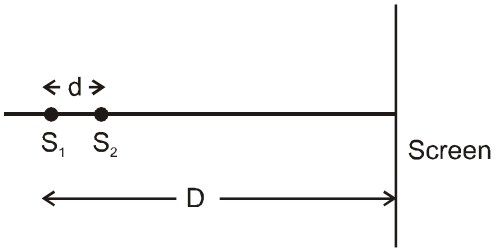
5. At two points P and Q on a screen in Young's double slit experiment, waves from slits S_1 and S_2 have a path difference of 0 and $\lambda/4$ respectively. The ratio of intensities at P and Q will be :
[AIEEE 2011, 11 May; 4/120, –1]
(1) 2 : 1 (2) $\sqrt{2}$: 1 (3) 4 : 1 (4) 3 : 2

6. In a Young's double slit experiment, the two slits act as coherent sources of waves of equal amplitude A and wavelength λ. In another experiment with the same arrangement the two slits are made to act as incoherent sources of waves of same amplitude and wavelength. If the intensity at the middle point of the screen in the first case is I_1 and in the second case is I_2, then the ratio $\frac{I_1}{I_2}$ is :
[AIEEE 2011, 11 May ; 4/120, –1]
(1) 2 (2) 1 (3) 0.5 (4) 4

7. **Statement-1 :** On viewing the clear blue portion of the sky through a Calcite Crystal, the intensity of transmitted light varies as the crystal is rotated.
Statement-2 : The light coming from the sky is polarized due to scattering of sun light by particles in the atmosphere. The scattering is largest for blue light **[AIEEE 2011, 11 May; 4/120, –1]**
(1) Statement-1 is true, statement-2 is false.
(2) Statement-1 is true, statement-2 is true, statement-2 is the correct explanation of statment-1
(3) Statement-1 is true, statement-2 is true, statement-2 is not the correct explanation of statement-1
(4) Statement-1 is false, statement-2 is true.

8. **Direction :** The question has a paragraph followed by two statements, Statement-1 and Statement-2. Of the given four alternatives after the statements, choose the one that describes the statements.
A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plate. With monochromatic light, this film gives an interference pattern due to light reflected from the top (convex) surface and the bottom (glass plate) surface of the film. **[AIEEE - 2011, 4/120, –1]**
Statement-1 : When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of π
Statement-2 : The centre of the interference pattern is dark.
(1) Statement-1 is true, statement-2 is false.
(2) Statement-1 is true, Statement-2 is true, Statement-2 is the correct explanation of Statement-1
(3) Statement-1 is true, Statement-2 is true, Statement-2 is not the correct explanation of Statement-1
(4) Statement-1 is false, Statement-2 is true

9. Two coherent point sources S_1 and S_2 are separated by a small distance 'd' as shown. The fringes obtained on the screen will be : **[JEE (Main) 2013, 4/120, –1]**



(1) points (2) straight lines (3) semi-circles (4) concentric circles

10. Tow beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30º makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals **[JEE (Main) 2014; 4/120, –1]**

(1) 3 (2) $\frac{3}{2}$ (3) 1 (4) $\frac{1}{3}$

11. On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to conclude that as it travels, the light beam : **[JEE (Main) 2015; 4/120, –1]**

(1) becomes narrower (2) goes horizontally without any deflection
(3) bends downwards (4) bends upwards

12. Assuming human pupil to have a radius of 0.25 cm and a comfortable viewing distance of 25 cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is : **[JEE (Main) 2015; 4/120, –1]**

(1) 1 μm (2) 30 μm (3) 100 μm (4) 300 μm

13. The box of a pin hole camera, of length L, has hole of radius a. it is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say b_{min}) when : **[JEE (Main) 2016; 4/120, –1]**

(1) $a = \sqrt{\lambda L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$ (2) $a = \sqrt{\lambda L}$ and $\sqrt{4\lambda L}$

(3) $a = \frac{\lambda^2}{L}$ and $b_{min} = \sqrt{4\lambda L}$ (4) $a = \frac{\lambda^2}{L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$

14. In a Young's double slit experiment, slits are separated by 0.5 mm, and the screen is placed 150 cm away. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is : **[JEE (Main) 2017 ; 4/120, –1]**

(1) 15.6 mm (2) 1.56 mm (3) 7.8 mm (4) 9.75 mm

15. Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is : **[JEE (Main) 2018 ; 4/120, –1]**

(1) 45° (2) 60° (3) 0° (4) 30°

16. The angular width of the central maximum in a single slit diffraction pattern is 60°. The width of the slit is 1 μm. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? (i.e. distance between the centres of each slit.) **[JEE (Main) 2018; 4/120, –1]**

(1) 75 μm (2) 100 μm (3) 25 μm (4) 50 μm

Exercise-1

EXERCISE-1

PART-I

SECTION (A) :

A-1. (a) $9I$ (b) $5I$ (c) I

A-2. 2

SECTION (B) :

B-1. (a) Angular separation of the fringes remains constant ($= \lambda / d$).The actual separation of the fringe increases in proportion to the distance of the screen from the plane of the two slits.

(b) The separation of the fringes (and also angular separation) decrease.

(c) The separation of the fringe (and also angular separation) decreases.

(d) By slightly increasing the width of the slits, we are only increasing the intensity of incident beam. Again no change in λ, D, d. so β unchanged but sharpness of the fringe increase.

B-2. 1.625 mm **B-3.** 0.30 mm

B-4. $\frac{180}{\pi} \times 2 \times 10^{-4}$ degree = 0.011º

SECTION (C) :

C-1. 0.9 mm

SECTION (D) :

D-1. (a) $\beta = 4.0 \times 10^{-4}$ m (b) $\frac{\beta}{3}$ and $\frac{2\beta}{3}$

D-2. $\frac{\lambda}{4(\mu-1)}$

SECTION (E) :

E-1. Maximum

SECTION (F) :

F-1. 1.45 **F-2.** 100 nm

SECTION (G) :

G-1. (i) 1.8 μm (ii) 1.2 μm

PART-II

SECTION (A) :

A-1. (A) **A-2.** (C)

SECTION (B) :

B-1. (C) **B-2.** (D) **B-3.** (A)

B-4. (B) **B-5.** (D) **B-6.** (C)

B-7. (D)

SECTION (C) :

C-1. (D) **C-2.** (C)

SECTION (D) :

D-1. (A) **D-2.** (D)

SECTION (E) :

E-1. (D) **E-2.** (A)

SECTION (F) :

F-1. (B)

SECTION (G) :

G-1. (D) **G-2.** (A) **G-3.** (A)

G-4. (B) **G-5.** (A) **G-6.** (C)

G-7. (B) **G-8.** (C) **G-9.** (D)

PART-III

1. (A)– q,r,s ; (B)–p,q,r,s ; (C)–q,r,s ; (D) – p,q,r,s

2. (A) – r, s;(B)–p, q, s; (C) – p, q, s ; (D) – r, s

EXERCISE-2

PART-I

1. (B) **2.** (C) **3.** (D)

4. (A) **5.** (A) **6.** (C)

7. (B) **8.** (B) **9.** (B)

10. (D) **11.** (C)

PART-II

1. 3 **2.** (i) 3 (ii) $3I$

3. 2 **4.** 49 **5.** 93

PART-III

1. (BD) **2.** (AC) **3.** (BCD)

4. (AC) **5.** (AC)

PART-IV

1. (D) **2.** (C) **3.** (B)

4. (A) **5.** (A) **6.** (C)

7. (D) **8.** (A)

EXERCISE-3

PART-I

1. (A) **2.** (C) **3.** (B)

4. (AB)

5. (A) $\rightarrow$ p, s; (B) $\rightarrow$ q; (C) $\rightarrow$ t ; (D) $\rightarrow$ r, s, t

6. (D) **7.** (B) **8.** (D)

9. (ABC) **10.** (3) **11.** (AC)

12. (AC)

PART-II

1. (4) **2.** (2) **3.** (4)

4. (1) **5.** (1) **6.** (1)

7. (2) **8.** (3) **9.** (4)

10. (4) **11.** (4) **12.** (2)

13. (2) **14.** (3) **15.** (1)

16. (3)